(3) Evaluate the given game tree using the Mini-max Algorithm and explain how Alpha-Beta pruning optimizes the search.

Indicate the pruned branches and determine the optimal utility value and decision for the Max player.

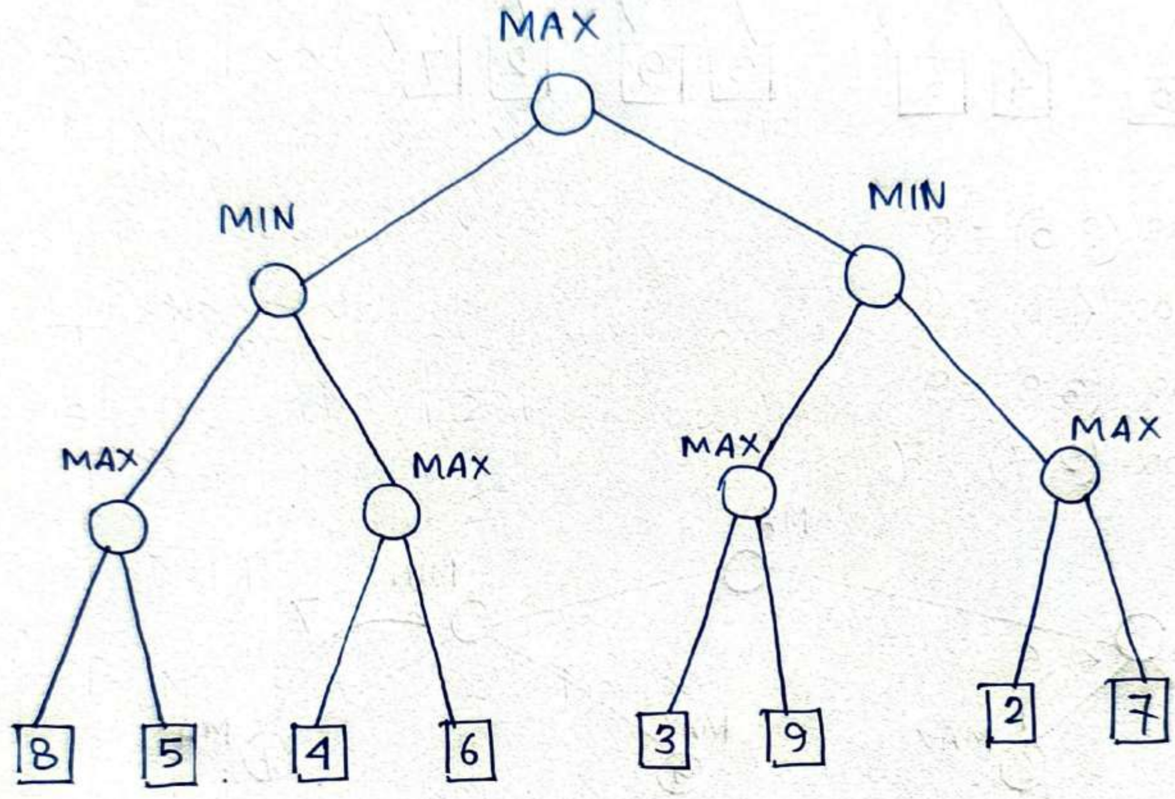
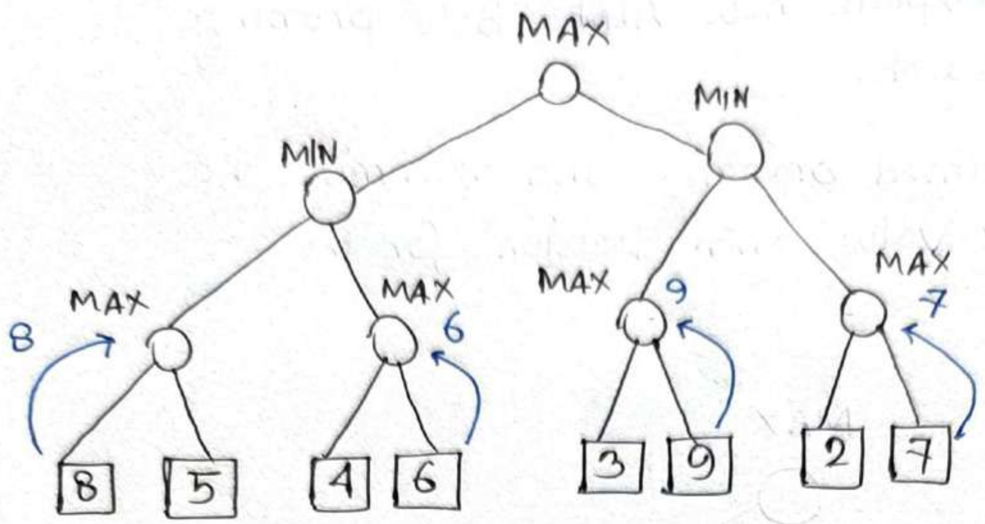


fig 1: Game Tree

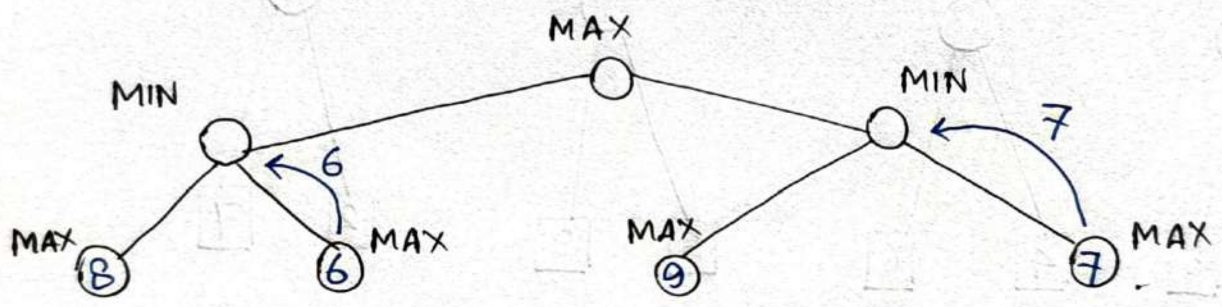
Answer:

In the given game tree, the Root Node is a MAX node. The next level contains MIN nodes. followed by MAX nodes and terminal utility values.

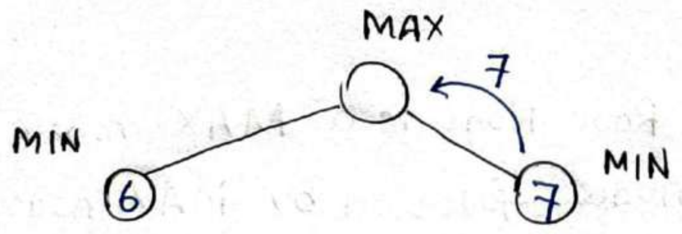
Applying Mini-max from the bottom up :



i) max(8,5) = 8
max(4,6) = 6
max (3,9) = 9
max (2,7) = 7



ii) min(8,6) = 6
min (9,7) = 7



iii) max (6,7) = 7

∴ The MAX player chooses the right subtree and the optimal utility value becomes 7.

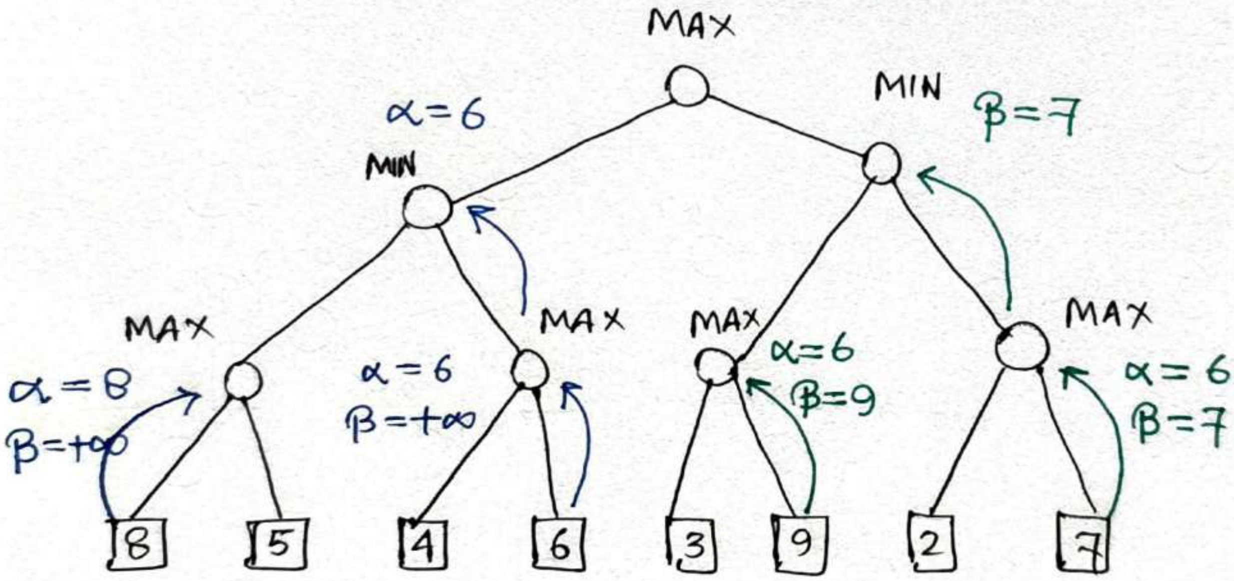
In the game tree, if the game has b possible moves at each Step and a depth of m, the complexity becomes $O(b^m)$
With Alpha-Beta pruning, the complexity improves to $O(b^{m/2})$

α is updated on MAX nodes,
β is updated on MIN nodes.

Pruning is performed if $\alpha \geq \beta$ occurs at any node.

Initially, $\alpha = -\infty$, $\beta = +\infty$

We evaluate the tree from left to right.



Now, $\alpha=6$, $\beta=+\infty$

The right sub tree is examined.

$\max(3,9)=9$

Now, $\alpha=6$, $\beta=9$

Since $\alpha < \beta$, no pruning occurs.

The second max child:

$$\max(2,7) = 7$$

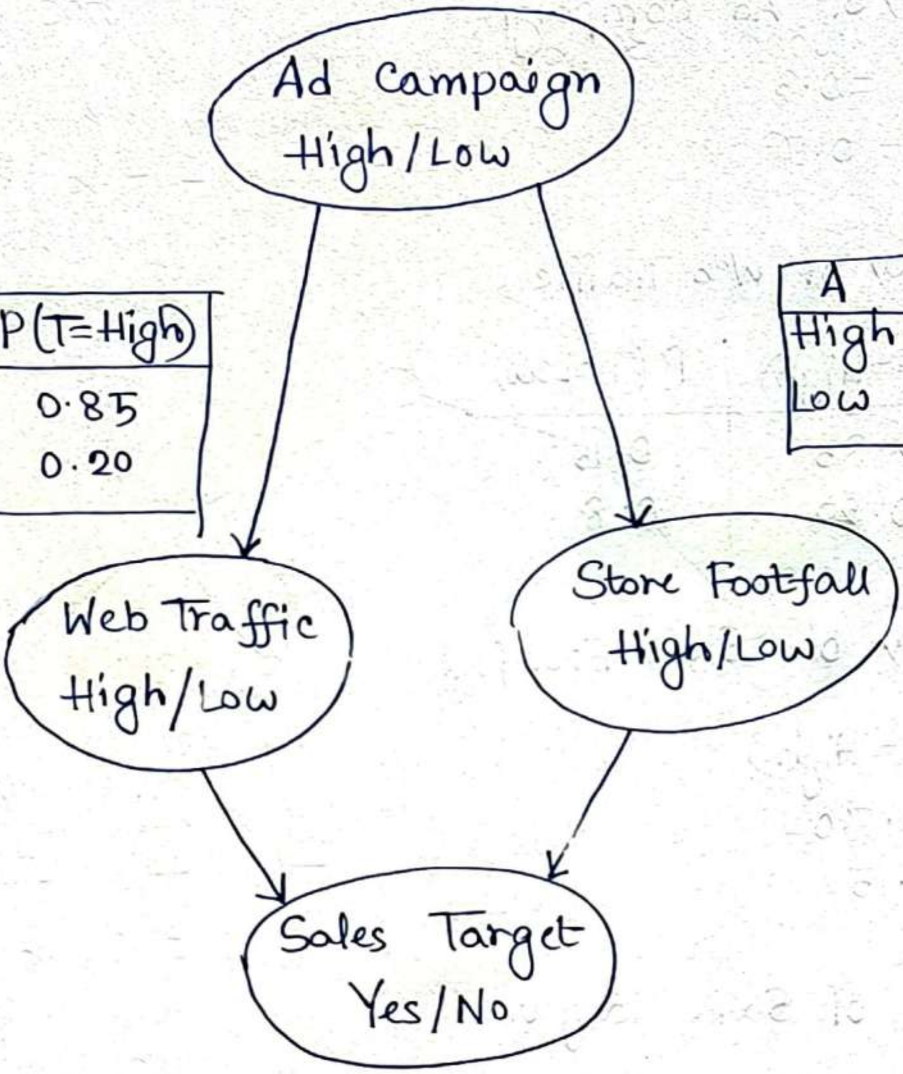
Now, $\alpha = 6$ and $\beta = 7$

Apparently, for the given tree and left to right evaluation order, the condition $\alpha \geq \beta$ never occurs.

Therefore, no branches are pruned.

Consider the Bayesian Network for the Marketing Campaign Problem. Calculate the probability that the Marketing Budget was High, given that we know the Sales Target was Not Met and that Store Footfall was also High.

$P(A = High) = 0.3$



A	P(T=High)
High	0.85
Low	0.20

A	P(F=High)
High	0.70
Low	0.15

T	F	P(S=Yes)
High	High	0.98
High	Low	0.60
Low	High	0.50
Low	Low	0.05

Answer :

From the given network,

A = Ad Campaign : High/Low

T = Web Traffic : High/Low

F = Store Foot-fall : High/Low

S = Sales Target : Yes/No

We have to find, $P(A \mid \neg S, F)$

Prior probability of Ad campaign,

$P(A = H) = 0.3$

$\therefore P(A = L) = 0.7$

Prior probability of Web Traffic,

A	P(T=High)	P(T=Low)
High	0.85	0.15
Low	0.20	0.8

Prior probability of Store Foot-fall,

A	P(T=High)
High	0.70
Low	0.15

Prior probability of Sales Target,

T	F	P(S=Yes)
High	High	0.98
High	Low	0.6
Low	High	0.5
Low	Low	0.05

However, our evidence is,

$S = No$

So, we need, $P(S=No \mid T, F)$

We know, F is High.

$\therefore$

T	F	P(S=No)
High	High	0.02
Low	High	0.5

If the buget was $\boxed{\text{High}}$, there are two possibilities for web traffic :

Case 1 : T = High

Case 2 : T = Low

For case 1 :

We get, $P(A=H, T=H, F=H, S=N)$

$\propto P(A=H)\, P(T=H \mid A=H)\, P(F=H \mid A=H)\, P(S=N \mid T=H, F=H)$

$= 0.3 \times 0.85 \times 0.7 \times 0.02$

$= 0.00357$

For case 2 :

We get $P(A=H, T=L, F=H, S=N)$

$\propto P(A=H)\, P(T=L \mid A=H)\, P(F=H \mid A=H)\, P(S=N \mid T=L, F=H)$

$= 0.3 \times 0.15 \times 0.7 \times 0.50$

$= 0.01575$

$\therefore$ So, all the probability associated with

A=H is : $0.00357 + 0.01575 = 0.01932$

If the budget was $\boxed{\text{LOW}}$ There are two cases for web traffic:

Case 1: T = High

Case 2: T = Low

For case 1, we get:

$$P(A=L, T=H, F=H, S=N)$$

$$\propto P(A=L)\,P(T=H \mid A=L)\,P(F=H \mid A=L)\,P(S=N \mid T=H, F=H)$$

$$= 0.7 \times 0.2 \times 0.15 \times 0.02$$

$$= 0.00042$$

For case 2, we get:

$$P(A=L, T=L, F=H, S=N)$$

$$\propto P(A=L)\,P(T=L \mid A=L)\,P(F=H \mid A=L)\,P(S=N \mid T=L, F=H)$$

$$= 0.7 \times 0.8 \times 0.15 \times 0.5$$

$$= 0.042$$

$\therefore$ All the possibilities associated with A=L is:

$$0.00042 + 0.042 = 0.04242$$

Normalizing for A=H,

$$P(A=H \mid S=N, F=H) = \frac{0.01932}{0.01932 + 0.04242}$$

$$= \boxed{0.313}$$